

Semester I exam practice

Paper 1 *[84 marks]*

1. A list of four physical quantities is

[1 mark]

- acceleration
- energy
- mass
- temperature

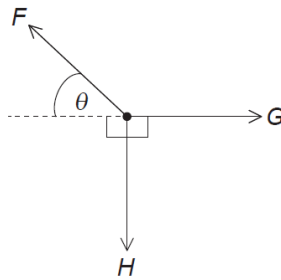
How many scalar quantities are in this list?

- A. 1
- B. 2
- C. 3
- D. 4

Markscheme

C

2. An object is held in equilibrium by three forces of magnitude F , G and H [1 mark] that act at a point in the same plane.



Three equations for these forces are

- I. $F \cos \theta = G$
- II. $F = G \cos \theta + H \sin \theta$
- III. $F = G + H$

Which equations are correct?

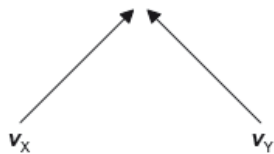
- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

Markscheme

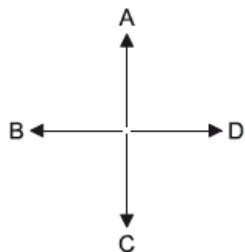
A

3. The velocities $\mathbf{v}_X$ and $\mathbf{v}_Y$ of two boats, X and Y, are shown.

[1 mark]



Which arrow represents the direction of the vector $\mathbf{v}_X - \mathbf{v}_Y$?



Markscheme

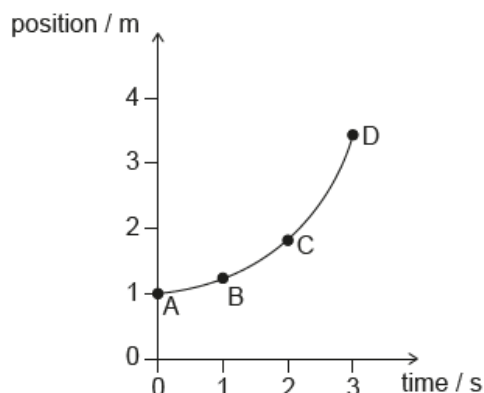
D

4. A large stone is dropped from a tall building. What is correct about the speed of the stone after 1 s? [1 mark]
- A. It is decreasing at increasing rate.
 - B. It is decreasing at decreasing rate.
 - C. It is increasing at increasing rate.
 - D. It is increasing at decreasing rate.

Markscheme

D

5. The graph shows how the position of an object varies with time in the interval from 0 to 3 s. [1 mark]



At which point does the instantaneous speed of the object equal its average speed over the interval from 0 to 3 s?

Markscheme

C

6. A car takes 20 minutes to climb a hill at constant speed. The mass of the car is 1200 kg and the car gains gravitational potential energy at a rate of 6.0 kW. Take the acceleration of gravity to be 10 m s^{-2} . What is the height of the hill? [1 mark]
- A. 0.6 m
 - B. 10 m
 - C. 600 m
 - D. 6000 m

Markscheme

C

7. A ball undergoes an elastic collision with a vertical wall. Which of the following is equal to zero? *[1 mark]*
- A. The change of the magnitude of linear momentum of the ball
 - B. The magnitude of the change of linear momentum of the ball
 - C. The rate of change of linear momentum of the ball
 - D. The impulse of the force on the ball

Markscheme

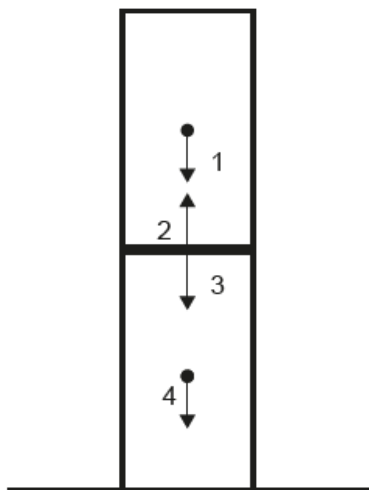
A

8. Two forces act on an object in different directions. The magnitudes of the forces are 18 N and 27 N. The mass of the object is 9.0 kg. What is a possible value for the acceleration of the object? *[1 mark]*
- A. 0 m s^{-2}
 - B. 0.5 m s^{-2}
 - C. 2.0 m s^{-2}
 - D. 6.0 m s^{-2}

Markscheme

C

9. Two identical boxes are stored in a warehouse as shown in the diagram. [1 mark]
Two forces acting on the top box and two forces acting on the bottom box are shown.



Which is a force pair according to Newton's third law?

- A. 1 and 2
- B. 3 and 4
- C. 2 and 3
- D. 2 and 4

Markscheme

C

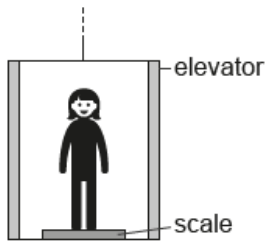
10. A person is standing at rest on the ground and experiences a downward gravitational force W and an upward normal force from the ground N . Which, according to Newton's third law, is the force that together with W forms a force pair? [1 mark]
- A. The gravitational force W acting upwards on the ground.
 - B. The gravitational force W acting upwards on the person.
 - C. The normal force N acting upwards on the person.
 - D. The normal force N acting downwards on the ground.

Markscheme

A

11. A person with a weight of 600 N stands on a scale in an elevator.

[1 mark]



What is the acceleration of the elevator when the scale reads 900 N?

- A. 5.0 m s^{-2} downwards
- B. 1.5 m s^{-2} downwards
- C. 1.5 m s^{-2} upwards
- D. 5.0 m s^{-2} upwards

Markscheme

D

12. Two identical boxes containing different masses are sliding with the same [1 mark] initial speed on the same horizontal surface. They both come to rest under the influence of the frictional force of the surface. How do the frictional force and acceleration of the boxes compare?

	Frictional force	Acceleration
A.	different	different
B.	different	equal
C.	equal	different
D.	equal	equal

Markscheme

B

13. Two identical blocks, each of mass m and speed v , travel towards each other on a frictionless surface. [1 mark]



The blocks undergo a head-on collision. What is definitely true **immediately** after the collision?

- A. The momentum of each block is zero.
- B. The total momentum is zero.
- C. The momentum of each block is $2mv$.
- D. The total momentum is $2mv$.

Markscheme

B

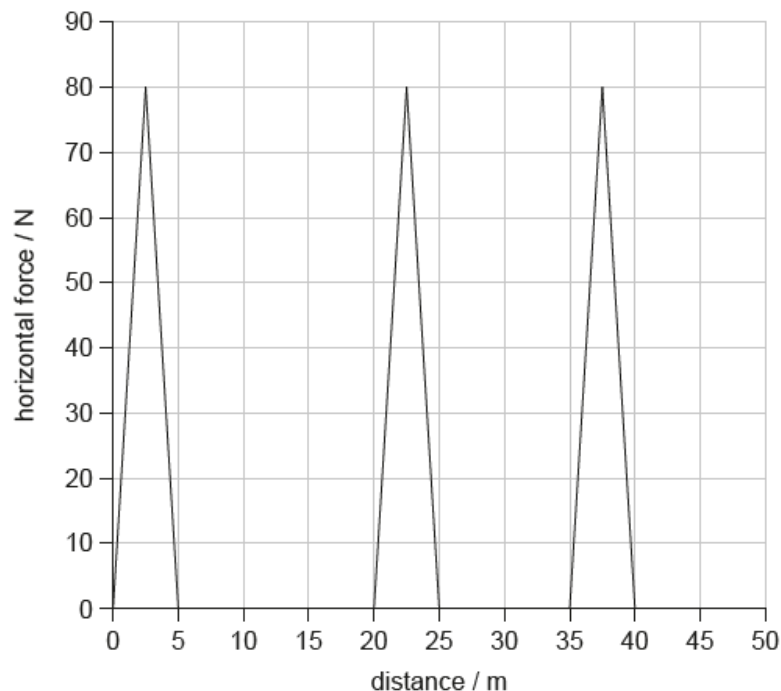
14. A projectile is launched upwards at an angle θ to the horizontal with an initial momentum p_0 and an initial energy E_0 . Air resistance is negligible. What are the momentum and total energy of the projectile at the highest point of the motion? [1 mark]

	Momentum	Energy
A.	$< p_0$	E_0
B.	p_0	E_0
C.	p_0	$< E_0$
D.	$< p_0$	$< E_0$

Markscheme

A

15. The graph shows the variation with distance of a horizontal force acting on an object. The object, initially at rest, moves horizontally through a distance of 50 m. [1 mark]



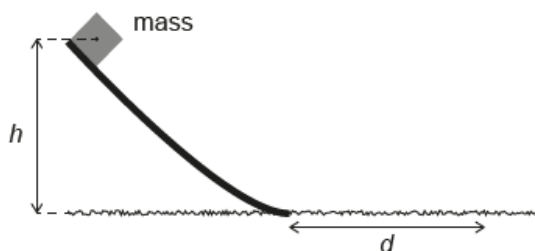
A constant frictional force of 2.0 N opposes the motion. What is the final kinetic energy of the object after it has moved 50 m?

- A. 100 J
- B. 500 J
- C. 600 J
- D. 1100 J

Markscheme

B

16. A mass is released from the top of a smooth ramp of height h . After leaving the ramp, the mass slides on a rough horizontal surface. [1 mark]



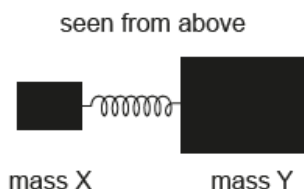
The mass comes to rest in a distance d . What is the coefficient of dynamic friction between the mass and the horizontal surface?

- A. $\frac{gd}{h}$
- B. $\sqrt{\frac{d}{2gh}}$
- C. $\frac{d}{h}$
- D. $\frac{h}{d}$

Markscheme

D

17. Masses X and Y rest on a smooth horizontal surface and are connected by [1 mark] a massless spring. The mass of X is 3.0 kg and the mass of Y is 6.0 kg. The masses are pushed toward each other until the elastic potential energy stored in the spring is 1.0 J.



The masses are released. What is the maximum speed reached by mass Y?

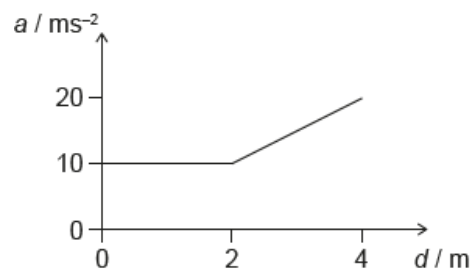
- A. 0.11 m s^{-1}
- B. 0.33 m s^{-1}
- C. 0.45 m s^{-1}
- D. 0.66 m s^{-1}

Markscheme

B

18. A force acts on an object of mass 40 kg. The graph shows how the acceleration a of the object varies with its displacement d .

[1 mark]



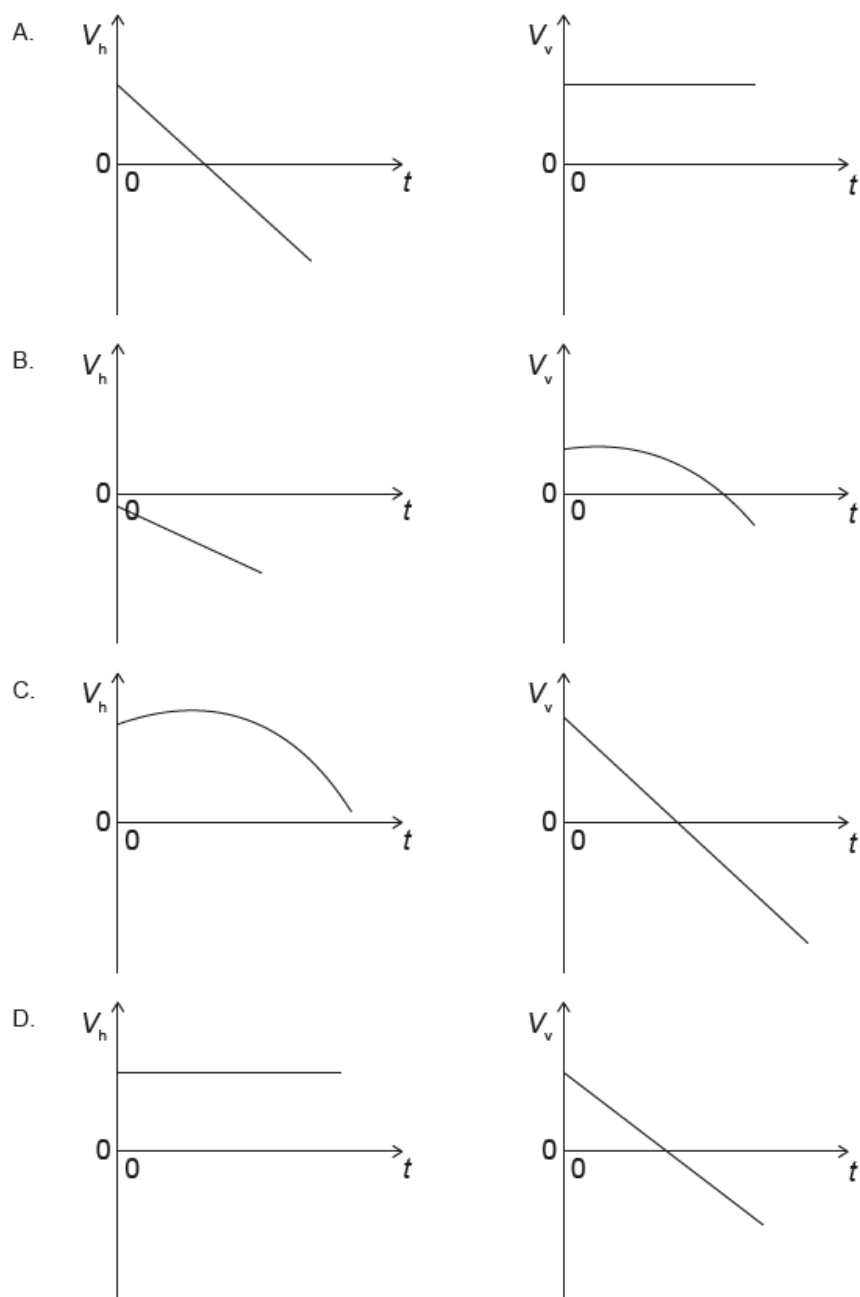
What is the work done by the force on the object?

- A. 50 J
- B. 2000 J
- C. 2400 J
- D. 3200 J

Markscheme

B

19. A projectile is launched at an angle above the horizontal with a horizontal [1 mark] component of velocity V_h and a vertical component of velocity V_v . Air resistance is negligible. Which graphs show the variation with time of V_h and of V_v ?



Markscheme

D

20. A horizontal force F acts on a sphere. A horizontal resistive force kv^2 acts [1 mark] on the sphere where v is the speed of the sphere and k is a constant. What is the terminal velocity of the sphere?

- A. $\sqrt{\frac{k}{F}}$
- B. $\frac{k}{F}$
- C. $\frac{F}{k}$
- D. $\sqrt{\frac{F}{k}}$

Markscheme

D

21. An object of mass 8.0 kg is falling vertically through the air. The drag [1 mark] force acting on the object is 60 N. What is the best estimate of the acceleration of the object?

- A. Zero
- B. 2.5 m s^{-2}
- C. 7.5 m s^{-2}
- D. 10 m s^{-2}

Markscheme

B

22. A balloon rises at a steady vertical velocity of 10 m s^{-1} . An object is [1 mark] dropped from the balloon at a height of 40 m above the ground. Air resistance is negligible. What is the time taken for the object to hit the ground?

- A. 10 s
- B. 5 s
- C. 4 s
- D. 2 s

Markscheme

C

23. An object of mass m strikes a vertical wall horizontally at speed U . The object rebounds from the wall horizontally at speed V . [1 mark]

What is the magnitude of the change in the momentum of the object?

- A. 0
- B. $m(V - U)$
- C. $m(U - V)$
- D. $m(U + V)$

Markscheme

D

24. A body is held in translational equilibrium by three coplanar forces of magnitude 3 N, 4 N and 5 N. Three statements about these forces are [1 mark]

- I. all forces are perpendicular to each other
- II. the forces cannot act in the same direction
- III. the vector sum of the forces is equal to zero.

Which statements are true?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

Markscheme

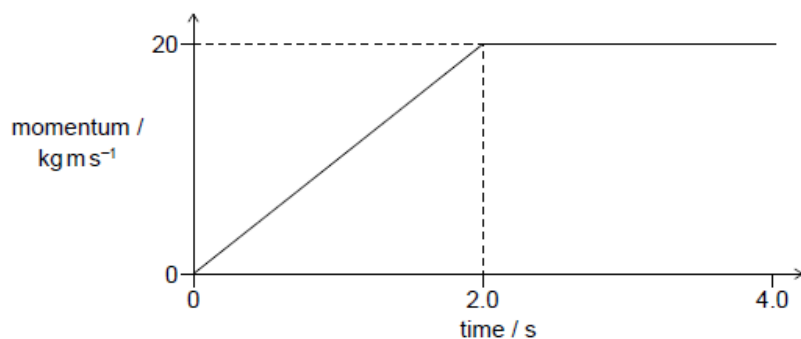
C

25. A car is driven from rest along a straight horizontal road. The car engine exerts a constant driving force. Friction and air resistance are negligible. How does the power developed by the engine change with the distance travelled? [1 mark]
- A. Power does not change.
 - B. Power decreases linearly.
 - C. Power increases linearly.
 - D. Power increases non-linearly.

Markscheme

D

26. The graph shows the variation of momentum with time for an object. [1 mark]



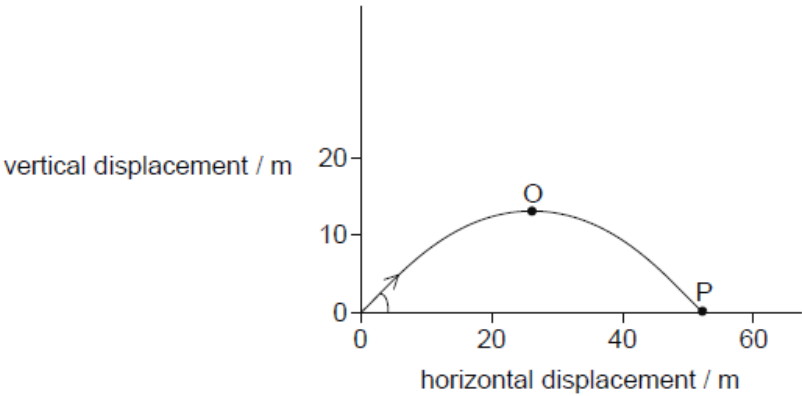
What net force acts on the object for the first 2.0 s and for the second 2.0 s of the motion?

	Force for first 2.0 s / N	Force for second 2.0 s / N
A.	10	0
B.	20	40
C.	10	40
D.	20	0

Markscheme

A

27. A projectile is fired at an angle to the horizontal. Air resistance is negligible. The path of the projectile is shown.
- [1 mark]



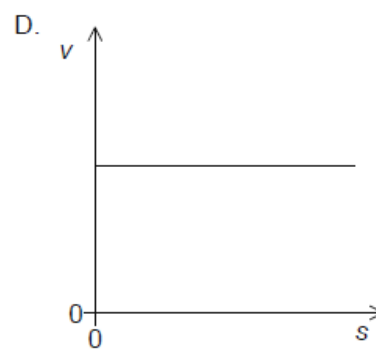
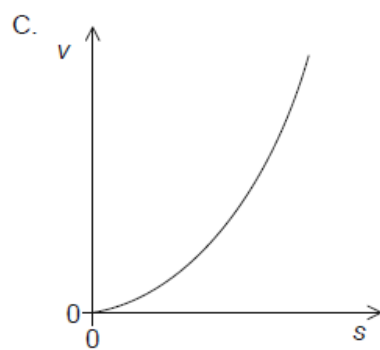
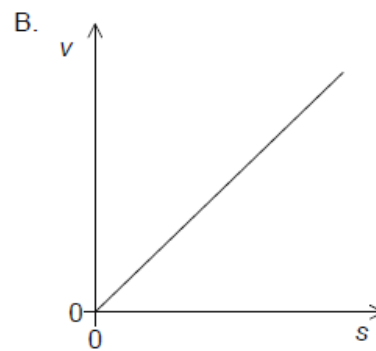
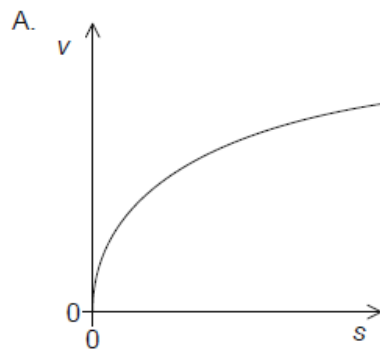
Which gives the magnitude of the horizontal component and the magnitude of the vertical component of the velocity of the projectile between O and P?

	Magnitude of horizontal component of velocity	Magnitude of vertical component of velocity
A.	constant	increases
B.	constant	constant
C.	increases	increases
D.	increases	constant

Markscheme

A

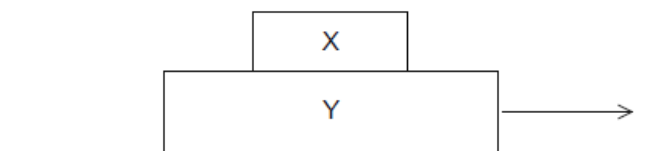
28. A runner starts from rest and accelerates at a constant rate throughout a [1 mark]
race. Which graph shows the variation of speed v of the runner with
distance travelled s ?



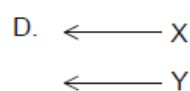
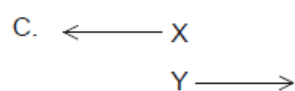
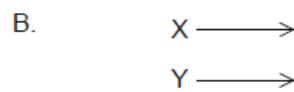
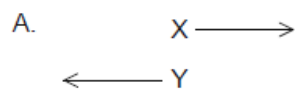
Markscheme

A

29. Two blocks X and Y rest on a frictionless horizontal surface as shown. A [1 mark]
horizontal force is now applied to the larger block and the two blocks
move together with the same speed and acceleration.



Which free-body diagram shows the frictional forces between the two blocks?

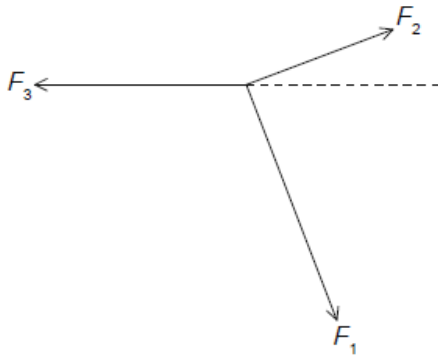


Markscheme

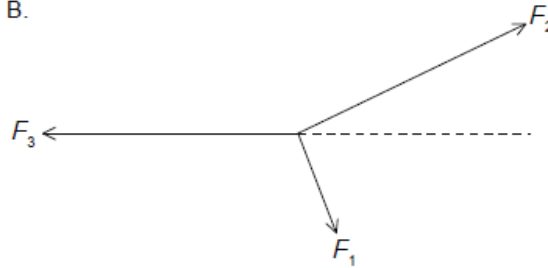
A

30. Three forces act at a point. In which diagram is the point in equilibrium? [1 mark]

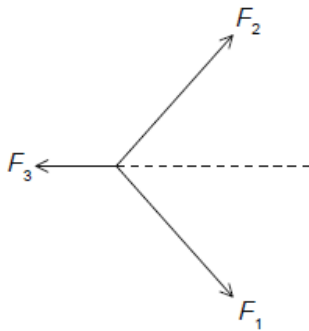
A.



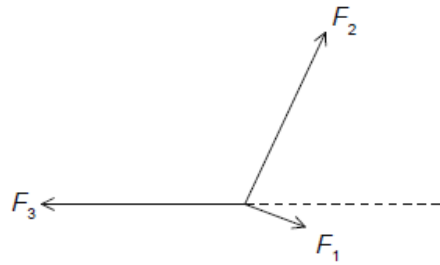
B.



C.



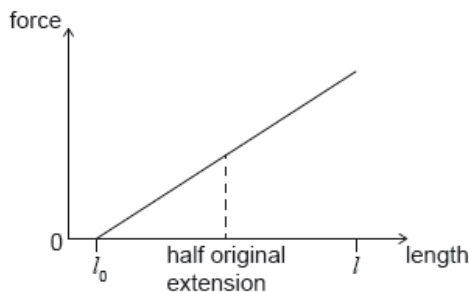
D.



Markscheme

B

31. An increasing force acts on a metal wire and the wire extends from an initial length l_0 to a new length l . The graph shows the variation of force with length for the wire. The energy required to extend the wire from l_0 to l is E . [1 mark]



The wire then contracts to half its original extension.

What is the work done by the wire as it contracts?

- A. $0.25E$
- B. $0.50E$
- C. $0.75E$
- D. E

Markscheme

C

32. The distances between successive positions of a moving car, measured at equal time intervals, are shown. [1 mark]



The car moves with

- A. acceleration that increases linearly with time.
- B. acceleration that increases non-linearly with time.
- C. constant speed.
- D. constant acceleration.

Markscheme

D

33. A motor of input power 160 W raises a mass of 8.0 kg vertically at a constant speed of 0.50 m s^{-1} .

[1 mark]

What is the efficiency of the system?

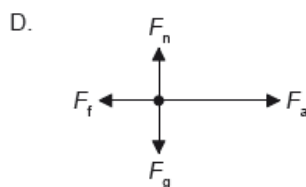
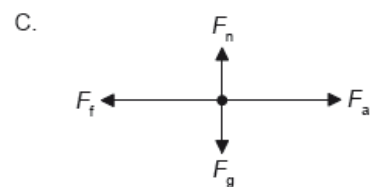
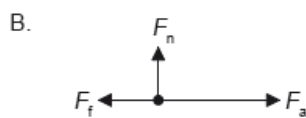
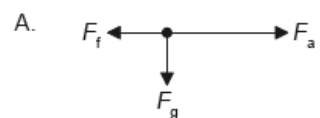
- A. 0.63%
- B. 25%
- C. 50%
- D. 100%

Markscheme

B

34. A box is accelerated to the right across rough ground by a horizontal force F_a . The force of friction is F_f . The weight of the box is F_g and the normal reaction is F_n . Which is the free-body diagram for this situation?

[1 mark]



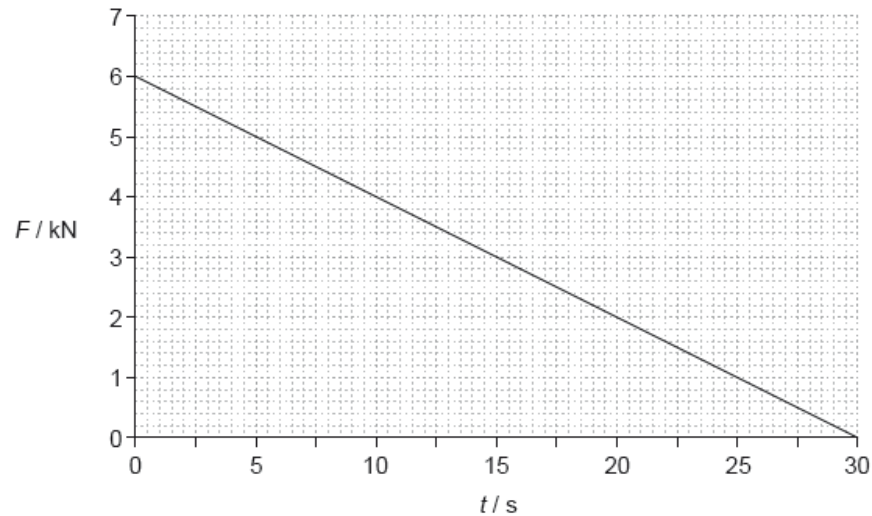
Markscheme

D

35. The graph shows the variation with time t of the force F acting on an object of mass 15 000 kg.

[1 mark]

The object is at rest at $t = 0$.



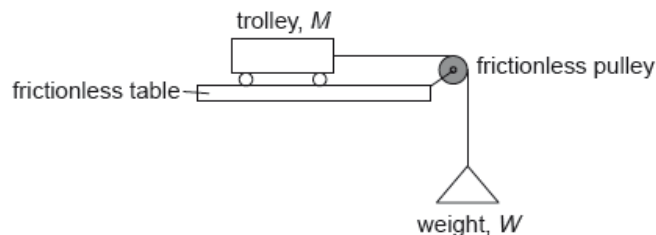
What is the speed of the object when $t = 30$ s?

- A. 0.18 m s^{-1}
- B. 6 m s^{-1}
- C. 12 m s^{-1}
- D. 180 m s^{-1}

Markscheme

B

36. A weight W is tied to a trolley of mass M by a light string passing over a frictionless pulley. The trolley has an acceleration a on a frictionless table. The acceleration due to gravity is g . [1 mark]



What is W ?

- A. $\frac{Mg}{(g-a)}$
- B. $\frac{Mg}{(g+a)}$
- C. $\frac{Ma}{(g-a)}$
- D. $\frac{Ma}{(g+a)}$

Markscheme

A

37. A parachutist of total mass 70 kg is falling vertically through the air at a constant speed of 8 m s^{-1} . [1 mark]

What is the total upward force acting on the parachutist?

- A. 0 N
- B. 70 N
- C. 560 N
- D. 700 N

Markscheme

D

38. A stopper of mass 8 g leaves the opening of a container that contains pressurized gas. The stopper accelerates from rest for a time of 16 ms and leaves the container at a speed of 20 m s^{-1} . [1 mark]

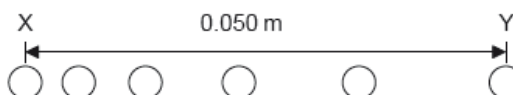
What is the order of magnitude of the force acting on the stopper?

- A. 10^{-3} N
- B. 10^0 N
- C. 10^1 N
- D. 10^3 N

Markscheme

C

39. A ball starts from rest and moves horizontally. Six positions of the ball are shown at time intervals of 1.0 ms. The horizontal distance between X, the initial position, and Y, the final position, is 0.050 m. [1 mark]



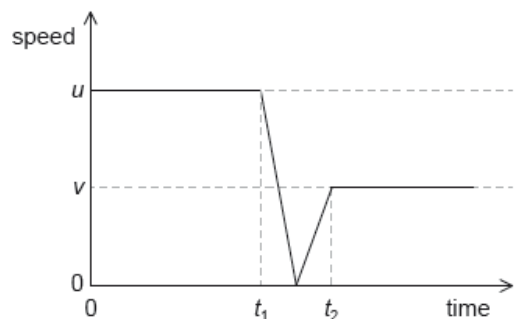
What is the average acceleration of the ball between X and Y?

- A. 2000 m s^{-2}
- B. 4000 m s^{-2}
- C. 5000 m s^{-2}
- D. 8000 m s^{-2}

Markscheme

B

40. A ball of mass m collides with a vertical wall with an initial horizontal speed u and rebounds with a horizontal speed v . The graph shows the variation of the speed of the ball with time. [1 mark]



What is the magnitude of the mean net force on the ball during the collision?

- A. $\frac{m(u-v)}{(t_2+t_1)}$
B. $\frac{m(u-v)}{(t_2-t_1)}$
C. $\frac{m(u+v)}{(t_2+t_1)}$
D. $\frac{m(u+v)}{(t_2-t_1)}$

Markscheme

D

41. The variation of the displacement of an object with time is shown on a graph. What does the area under the graph represent? [1 mark]

- A. No physical quantity
B. Velocity
C. Acceleration
D. Impulse

Markscheme

A

42. Which aspect of thermal physics is best explained by the molecular kinetic model? [1 mark]
- A. The equation of state of ideal gases
 - B. The difference between Celsius and Kelvin temperature
 - C. The value of the Avogadro constant
 - D. The existence of gaseous isotopes

Markscheme

A

43. When 40 kJ of energy is transferred to a quantity of a liquid substance, its temperature increases by 20 K. When 600 kJ of energy is transferred to the same quantity of the liquid at its boiling temperature, it vaporizes completely at constant temperature. What is [1 mark]
- $$\frac{\text{specific latent heat of vaporization}}{\text{specific heat capacity of the liquid}}$$
- for this substance?
- A. 15 K^{-1}
 - B. 15 K
 - C. 300 K^{-1}
 - D. 300 K

Markscheme

D

44. A quantity of 2.00 mol of an ideal gas is maintained at a temperature of 127°C in a container of volume 0.083 m^3 . What is the pressure of the gas? [1 mark]
- A. 8 kPa
 - B. 25 kPa
 - C. 40 kPa
 - D. 80 kPa

Markscheme

D

45. What is the relation between the value of the unified atomic mass unit in grams and the value of Avogadro's constant in mol^{-1} ? [1 mark]
- A. Their ratio is 1.
 - B. Their product is 1.
 - C. Their sum is 1.
 - D. Their difference is 0.

Markscheme

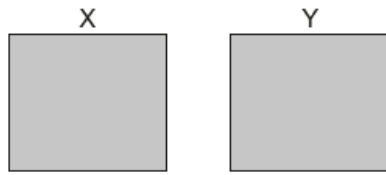
B

46. A sample of oxygen gas with a volume of 3.0 m^3 is at 100°C . The gas is heated so that it expands at a constant pressure to a final volume of 6.0 m^3 . What is the final temperature of the gas? [1 mark]
- A. 750°C
 - B. 470°C
 - C. 370°C
 - D. 200°C

Markscheme

B

47. Two identical containers X and Y each contain an ideal gas. X has N molecules of gas at an absolute temperature of T and Y has $3N$ molecules of gas at an absolute temperature of $\frac{T}{2}$. What is the ratio of the pressures $\frac{P_Y}{P_X}$? [1 mark]



- A. $\frac{1}{6}$
- B. $\frac{2}{3}$
- C. $\frac{3}{2}$
- D. 6

Markscheme

C

48. A piece of metal at a temperature of 100°C is dropped into an equal mass of water at a temperature of 15°C in a container of negligible mass. The specific heat capacity of water is four times that of the metal. What is the final temperature of the mixture? [1 mark]

- A. 83°C
- B. 57°C
- C. 45°C
- D. 32°C

Markscheme

D

49. Two ideal gases X and Y are at the same temperature. The mass of a particle of gas X is larger than the mass of a particle of gas Y. Which is correct about the average kinetic energy and the average speed of the particles in gases X and Y? [1 mark]

	Average kinetic energy	Average speed
A.	larger for Y	larger for Y
B.	same	larger for Y
C.	same	same
D.	larger for Y	same

Markscheme

B

50. An ideal gas of constant mass is heated in a container of constant volume. [1 mark]

What is the reason for the increase in pressure of the gas?

- A. The average number of molecules per unit volume increases.
- B. The average force per impact at the container wall increases.
- C. Molecules collide with each other more frequently.
- D. Molecules occupy a greater fractional volume of the container.

Markscheme

B

51. A bicycle of mass M comes to rest from speed v using the back brake. The brake has a specific heat capacity of c and a mass m . Half of the kinetic energy is absorbed by the brake. [1 mark]

What is the change in temperature of the brake?

- A. $\frac{Mv^2}{4mc}$
- B. $\frac{Mv^2}{2mc}$
- C. $\frac{mv^2}{4Mc}$
- D. $\frac{mv^2}{2Mc}$

Markscheme

A

52. What is **not** an assumption of the kinetic model of an ideal gas? [1 mark]
- A. Attractive forces between molecules are negligible.
 - B. Collision duration is negligible compared with time between collisions.
 - C. Molecules suffer negligible momentum change during wall collisions.
 - D. Molecular volume is negligible compared with gas volume.

Markscheme

C

53. A mass m of water is at a temperature of 290 K. The specific heat capacity of water is c . Ice, at its melting point, is added to the water to reduce the water temperature to the freezing point. The specific latent heat of fusion for ice is L . What is the minimum mass of ice that is required? [1 mark]
- A. $\frac{17mc}{L}$
 - B. $\frac{290mc}{L}$
 - C. $\frac{17mL}{c}$
 - D. $\frac{290mL}{c}$

Markscheme

A

54. Energy is transferred to water in a flask at a rate P . The water reaches boiling point and then P is increased. What are the changes to the temperature of the water and to the rate of vaporization of the water after the change? [1 mark]

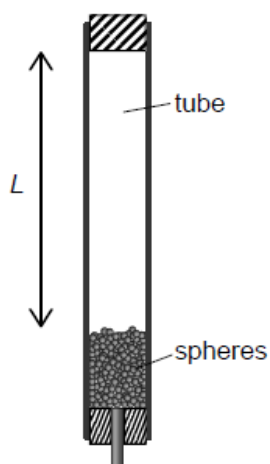
	Temperature	Rate of vaporization
A.	increases	unchanged
B.	increases	increases
C.	unchanged	unchanged
D.	unchanged	increases

Markscheme

D

55. An insulated tube is filled with a large number n of lead spheres, each of mass m . The tube is inverted s times so that the spheres completely fall through an average distance L each time. The temperature of the spheres is measured before and after the inversions and the resultant change in temperature is ΔT . [1 mark]

What is the specific heat capacity of lead?



- A. $\frac{sgL}{nm\Delta T}$
 B. $\frac{sgL}{\Delta T}$
 C. $\frac{sgL}{n\Delta T}$
 D. $\frac{gL}{m\Delta T}$

Markscheme

B

56. Boiling water is heated in a 2 kW electric kettle. The initial mass of water *[1 mark]* is 0.4 kg. Assume the specific latent heat of vaporization of water is 2 MJ kg⁻¹.

What is the time taken for all the water to vaporize?

- A. 250 s
- B. 400 s
- C. 2500 s
- D. 4000 s

Markscheme

B

57. The temperature of a fixed mass of an ideal gas changes from 200 °C to *[1 mark]* 400 °C.

What is $\frac{\text{mean kinetic energy of gas at } 200^\circ\text{C}}{\text{mean kinetic energy of gas at } 400^\circ\text{C}}$?

- A. 0.50
- B. 0.70
- C. 1.4
- D. 2.0

Markscheme

B

58. A container holds 20 g of argon-40 ($^{40}_{18}\text{Ar}$) and 40 g of neon-20 ($^{20}_{10}\text{Ne}$) . [1 mark]

What is $\frac{\text{number of atoms of argon -40}}{\text{number of atoms of neon -20}}$ in the container?

- A. 0.25
- B. 0.5
- C. 2
- D. 4

Markscheme

A

59. A 700 W electric heater is used to heat 1 kg of water without energy losses. The specific heat capacity of water is $4.2 \text{ kJ kg}^{-1} \text{ K}^{-1}$. What is the time taken to heat the water from 25°C to 95°C ? [1 mark]

- A. 7 s
- B. 30 s
- C. 7 minutes
- D. 420 minutes

Markscheme

C

60. Container X contains 1.0 mol of an ideal gas. Container Y contains 2.0 mol [1 mark] of the ideal gas. Y has four times the volume of X. The pressure in X is twice that in Y.

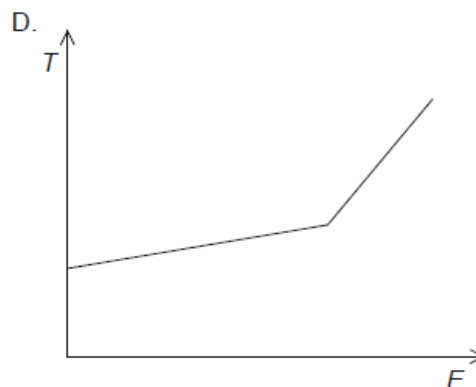
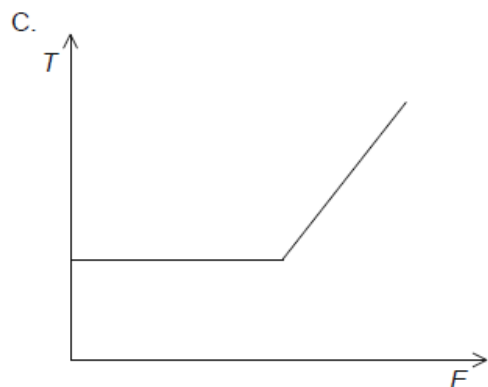
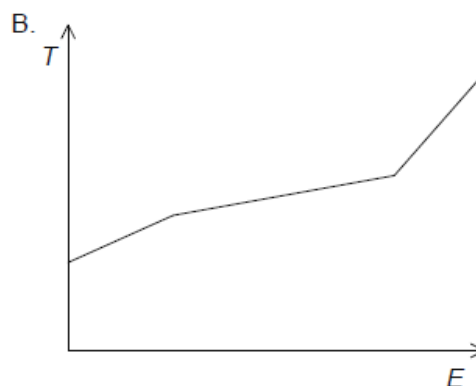
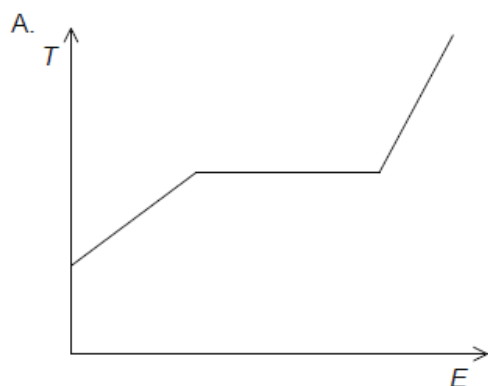
What is $\frac{\text{temperature of gas in X}}{\text{temperature of gas in Y}}$?

- A. $\frac{1}{4}$
- B. $\frac{1}{2}$
- C. 1
- D. 2

Markscheme

C

61. A solid substance has just reached its melting point. Thermal energy is supplied to the substance at a constant rate. Which graph shows the variation of the temperature T of the substance with energy E supplied? [1 mark]



Markscheme

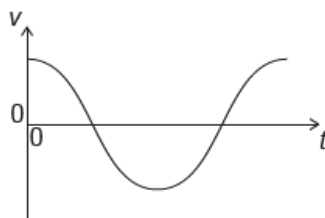
C

62. An ideal gas and a solid of the same substance are at the same temperature. The average kinetic energy of the gas molecules is E_g and the average kinetic energy of the solid molecules is E_s . What is the comparison between E_g and E_s ? [1 mark]
- A. E_g is less than E_s .
 - B. E_g equals E_s .
 - C. E_g is greater than E_s .
 - D. The relationship between E_g and E_s cannot be determined.

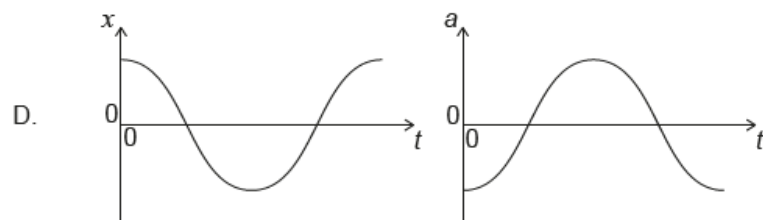
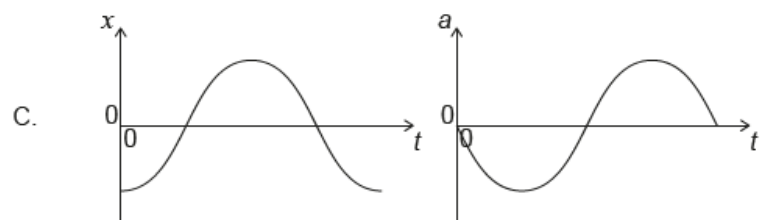
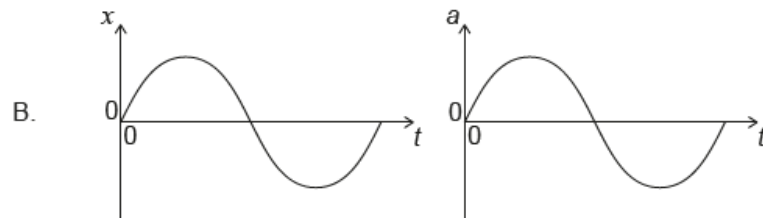
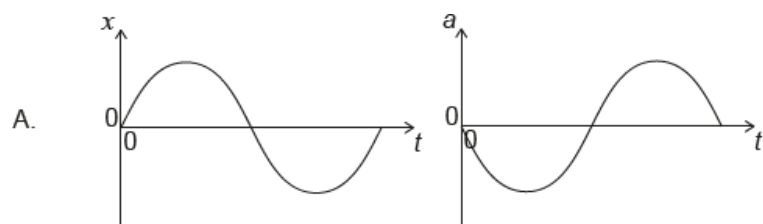
Markscheme

B

63. An object performs simple harmonic motion (shm). The graph shows how [1 mark]
the velocity v of the object varies with time t .



The displacement of the object is x and its acceleration is a . What is the variation of x with t and the variation of a with t ?



Markscheme

A

64. A sound wave has a frequency of 1.0 kHz and a wavelength of 0.33 m. [1 mark]
What is the distance travelled by the wave in 2.0 ms and the nature of the wave?

	Distance travelled in 2.0 ms	Nature of the wave
A.	0.17 m	longitudinal
B.	0.17 m	transverse
C.	0.66 m	longitudinal
D.	0.66 m	transverse

Markscheme

C

65. Two identical waves, each with amplitude X_0 and intensity I , interfere constructively. What are the amplitude and intensity of the resultant wave? [1 mark]

	Amplitude of the resultant wave	Intensity of the resultant wave
A.	X_0	$2I$
B.	$2X_0$	$2I$
C.	X_0	$4I$
D.	$2X_0$	$4I$

Markscheme

D

66. Three quantities used to describe a light wave are

[1 mark]

- I. frequency
- II. wavelength
- III. speed.

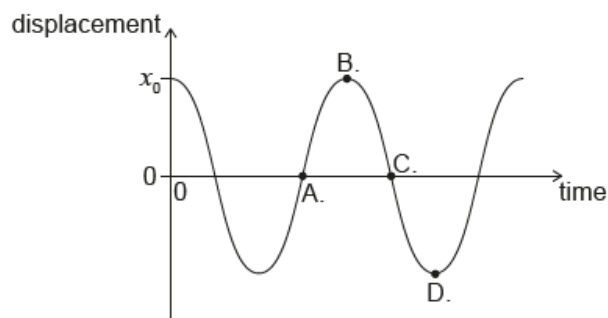
Which quantities increase when the light wave passes from water to air?

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

Markscheme

C

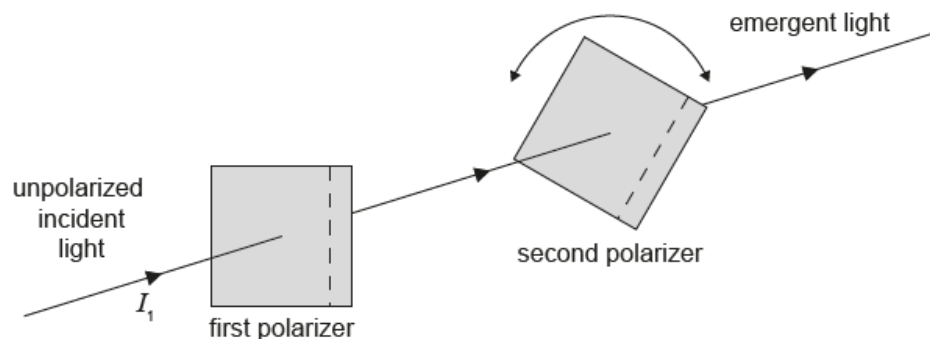
67. The bob of a pendulum has an initial displacement x_0 to the right. The bob is released and allowed to oscillate. The graph shows how the displacement varies with time. At which point is the velocity of the bob at its maximum magnitude directed towards the left? [1 mark]



Markscheme

C

68. Unpolarized light of intensity I_1 is incident on a polarizer. The light that passes through this polarizer then passes through a second polarizer. [1 mark]



The second polarizer can be rotated to vary the intensity of the emergent light. What is the maximum value of the intensity emerging from the second polarizer?

- A. $\frac{I_1}{4}$
- B. $\frac{I_1}{2}$
- C. $\frac{2I_1}{3}$
- D. I_1

Markscheme

B

69. A glass block has a refractive index in air of n_g . The glass block is placed in two different liquids: liquid X with a refractive index of n_X and liquid Y with a refractive index of n_Y . [1 mark]

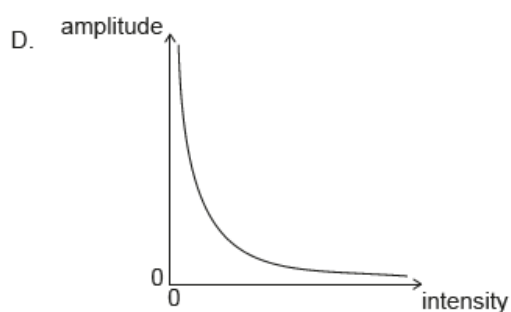
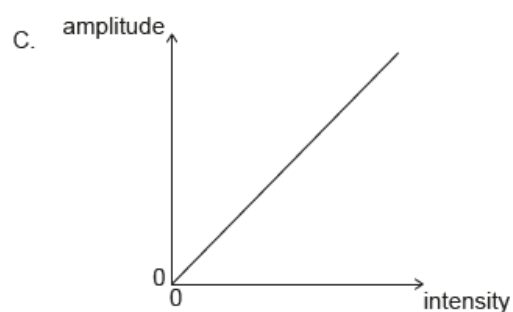
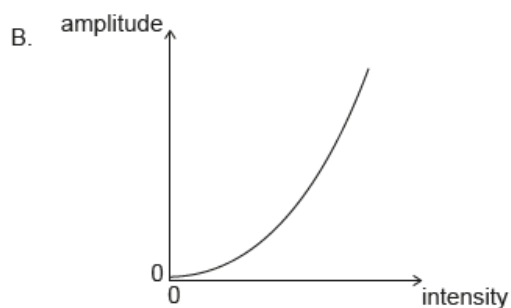
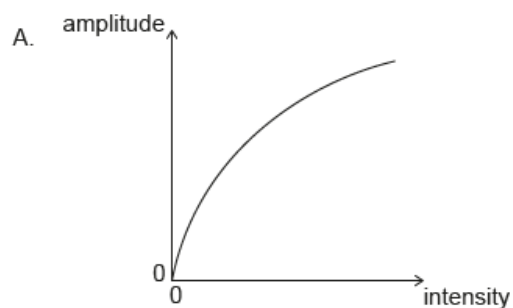
In liquid X $\frac{n_g}{n_X} = 2$ and in liquid Y $\frac{n_g}{n_Y} = 1.5$. What is $\frac{\text{speed of light in liquid X}}{\text{speed of light in liquid Y}}$?

- A. $\frac{2}{4}$
- B. $\frac{3}{4}$
- C. $\frac{4}{3}$
- D. 3

Markscheme

C

70. Which graph shows the variation of amplitude with intensity for a wave? [1 mark]



Markscheme

B

71. An object moves with simple harmonic motion. The acceleration of the object is [1 mark]

- A. constant.
- B. always directed away from the centre of the oscillation.
- C. a maximum at the centre of the oscillation.
- D. a maximum at the extremes of the oscillation.

Markscheme

D

72. What changes occur to the frequency and wavelength of monochromatic [1 mark]
light when it travels from glass to air?

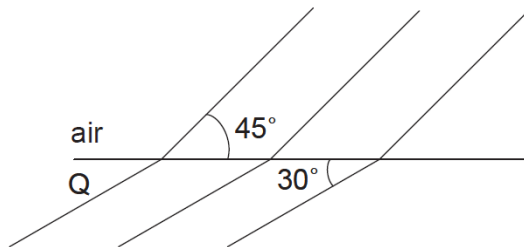
	Frequency	Wavelength
A.	stays the same	stays the same
B.	stays the same	increases
C.	increases	stays the same
D.	decreases	increases

Markscheme

B

73. Wavefronts travel from air to medium Q as shown.

[1 mark]



What is the refractive index of Q?

- A. $\frac{\sin 30^\circ}{\sin 45^\circ}$
- B. $\frac{\sin 45^\circ}{\sin 30^\circ}$
- C. $\frac{\sin 45^\circ}{\sin 60^\circ}$
- D. $\frac{\sin 60^\circ}{\sin 45^\circ}$

Markscheme

B

74. The motion of an object is described by the equation

[1 mark]

acceleration $\propto$ – displacement.

What is the direction of the acceleration relative to that of the displacement and what is the displacement when the speed is a maximum?

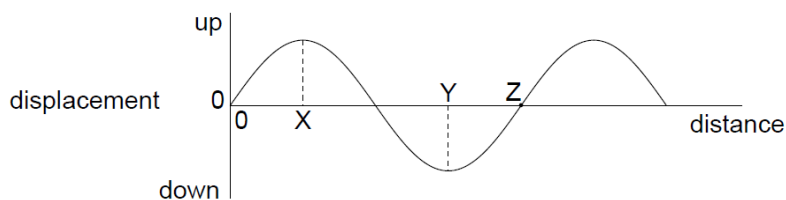
	Direction of acceleration relative to displacement	Displacement when speed is a maximum
A.	same	max
B.	same	zero
C.	opposite	max
D.	opposite	zero

Markscheme

D

75. A transverse travelling wave is moving through a medium. The graph shows, for one instant, the variation with distance of the displacement of particles in the medium.

[1 mark]



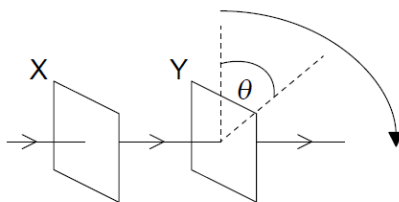
The frequency of the wave is 25 Hz and the speed of the wave is 100 m s^{-1} . What is correct for this wave?

- A. The particles at X and Y are in phase.
- B. The velocity of the particle at X is a maximum.
- C. The horizontal distance between X and Z is 3.0 m.
- D. The velocity of the particle at Y is 100 m s^{-1} .

Markscheme

C

76. Unpolarized light is incident on two polarizing filters X and Y. They are arranged so that light emerging from Y has a maximum intensity. X is fixed and Y is rotated through θ about the direction of the incident beam in its own plane. [1 mark]



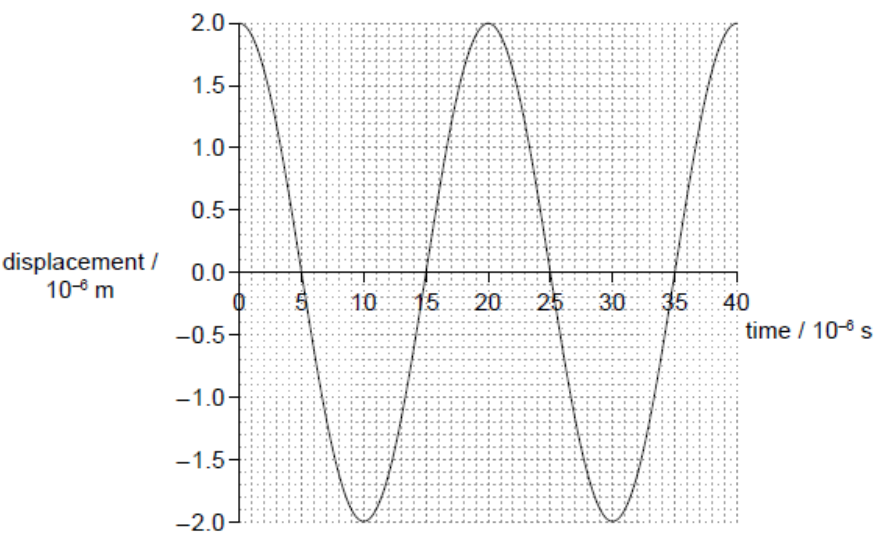
What are the first three successive values of θ for which the final transmitted intensity is a maximum?

- A. 90° , 180° , 270°
- B. 90° , 270° , 450°
- C. 180° , 360° , 540°
- D. 180° , 540° , 720°

Markscheme

C

77. The graph shows the variation with time for the displacement of a particle [1 mark] in a travelling wave.



What are the frequency and amplitude for the oscillation of the particle?

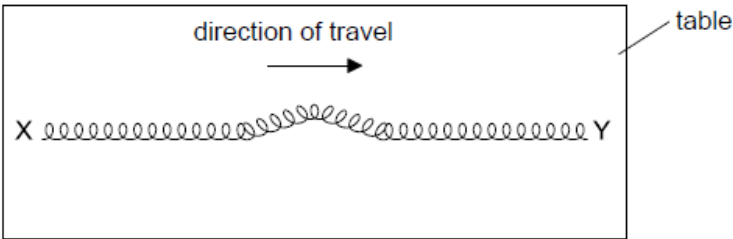
	Frequency / kHz	Amplitude / μm
A.	20	2
B.	20	4
C.	50	2
D.	50	4

Markscheme

C

78. A spring XY lies on a frictionless table with the end Y free.

[1 mark]



A horizontal pulse travels along the spring from X to Y. What happens when the pulse reaches Y?

- A. The pulse will be reflected towards X and inverted.
- B. The pulse will be reflected towards X and not be inverted.
- C. Y will move and the pulse will disappear.
- D. Y will not move and the pulse will disappear.

Markscheme

B

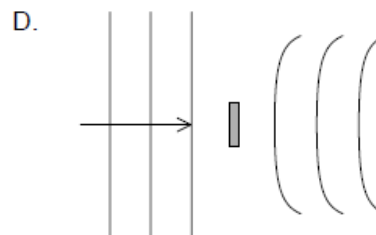
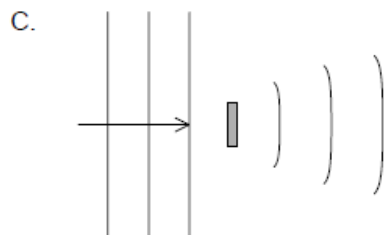
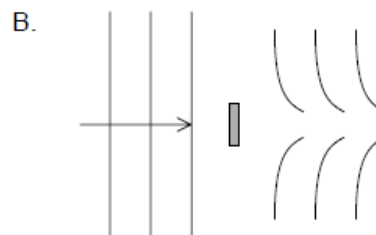
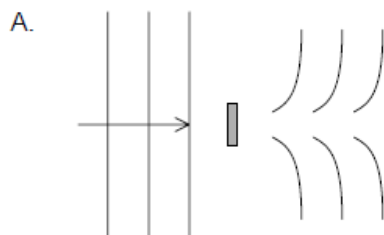
79. A point source of light of amplitude A_0 gives rise to a particular light intensity when viewed at a distance from the source. When the amplitude is increased and the viewing distance is doubled, the light intensity is doubled. What is the new amplitude of the source? [1 mark]

- A. $2A_0$
- B. $2\sqrt{2} A_0$
- C. $4A_0$
- D. $8A_0$

Markscheme

B

80. Which diagram shows the shape of the wavefront as a result of the diffraction of plane waves by an object? [1 mark]



Markscheme

A

81. A point source emits sound waves of amplitude A . The sound intensity at a distance d from the source is I . What is the sound intensity at a distance $0.5d$ from the source when the source emits waves of amplitude $2A$? [1 mark]

A. $16I$

B. $4I$

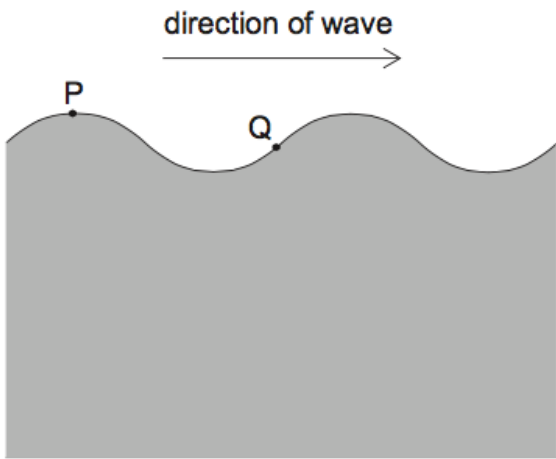
C. I

D. $\frac{1}{4}I$

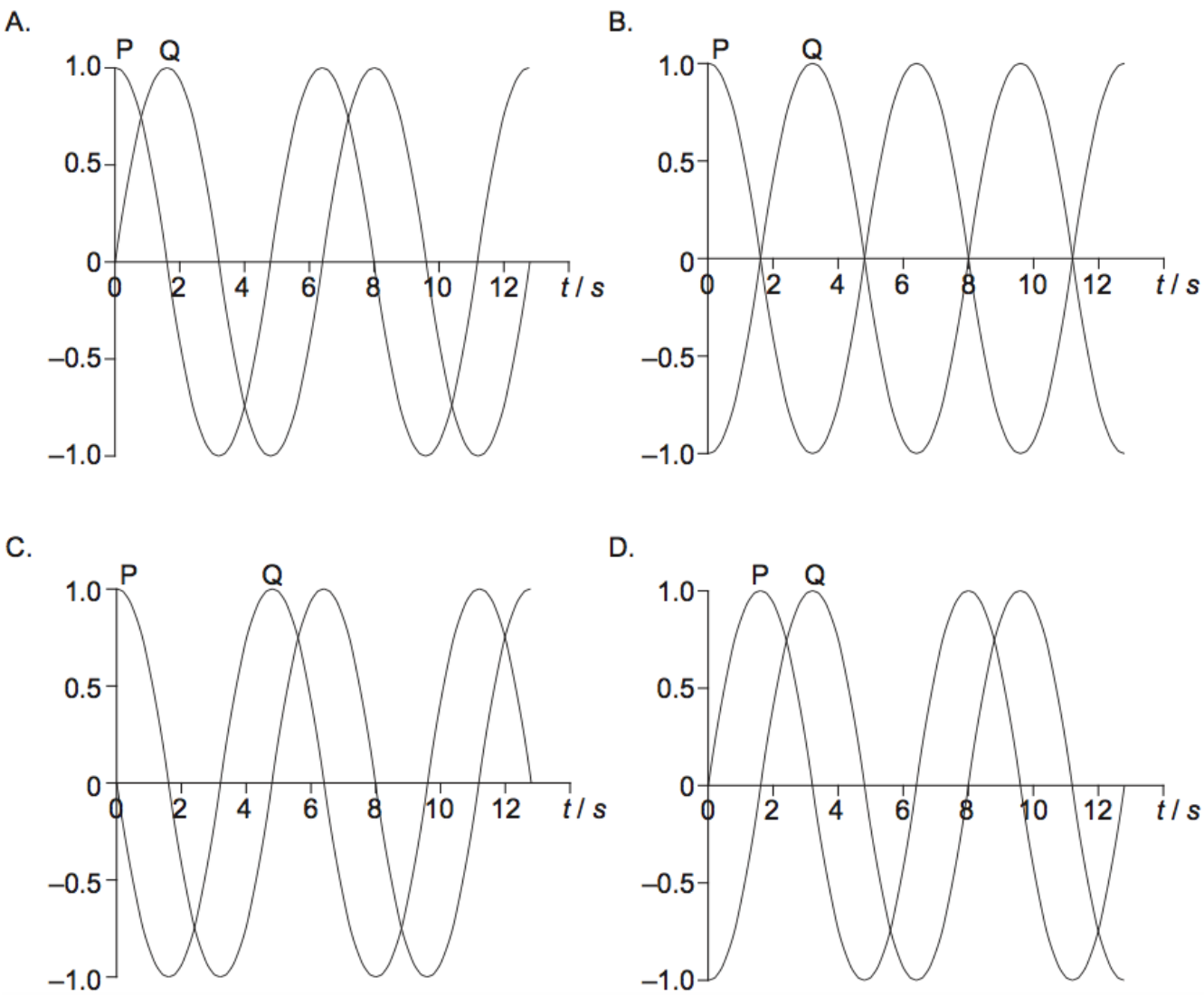
Markscheme

A

82. A water wave moves on the surface of a lake. P and Q are two points on the water surface. The wave is traveling towards the right. [1 mark]



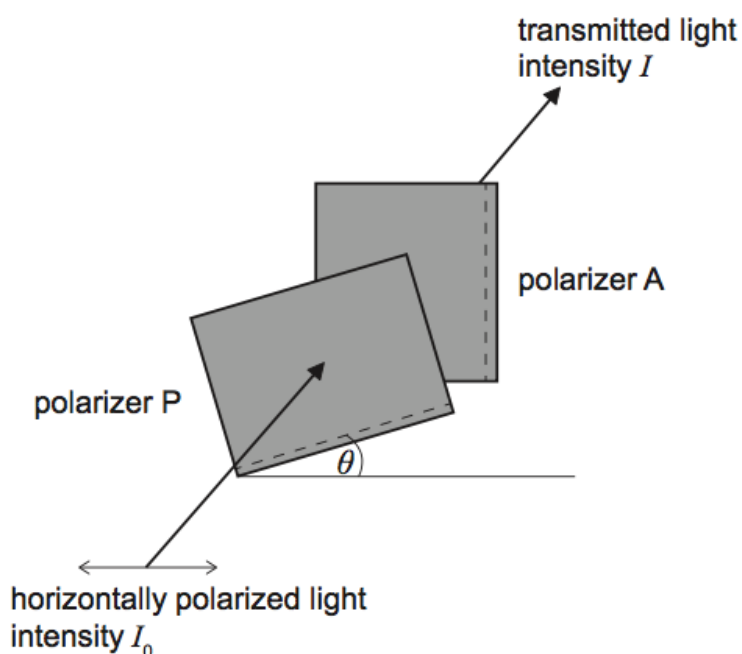
The diagram shows the wave at time $t = 0$. Which graph shows how the displacements of P and Q vary with t ?



Markscheme

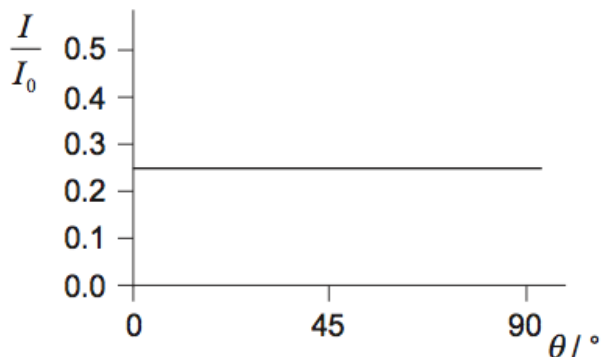
C

83. Horizontally polarized light of intensity I_0 enters a polarizer P whose polarization axis makes an angle of θ degrees with the horizontal. Light from P is then incident on a polarizer A with fixed vertical polarization axis. [1 mark]
 from P is then incident on a polarizer A with fixed vertical polarization axis.

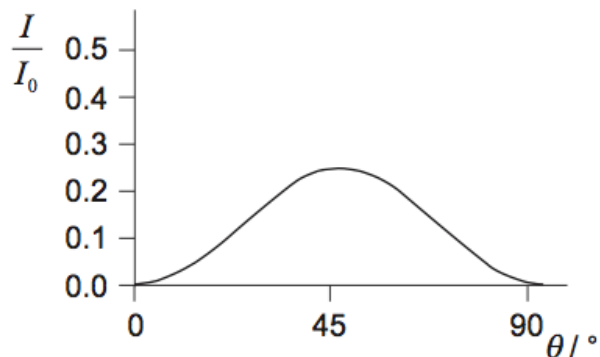


The angle θ is varied from 0 to 90 degrees. Which of the following represents the variation with θ of the intensity I of the light transmitted through A?

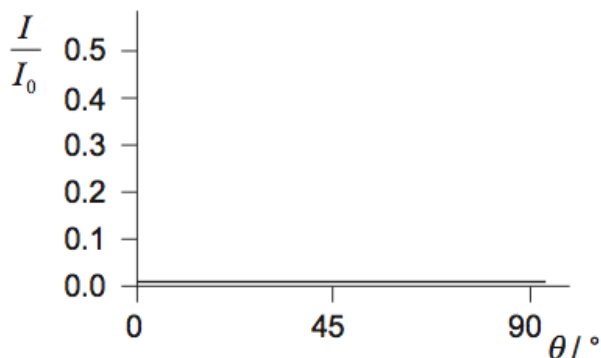
A.



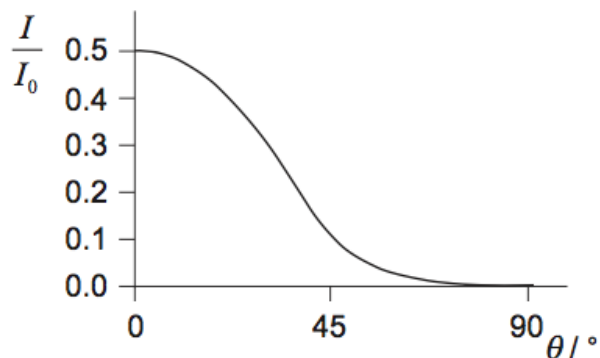
B.



C.



D.

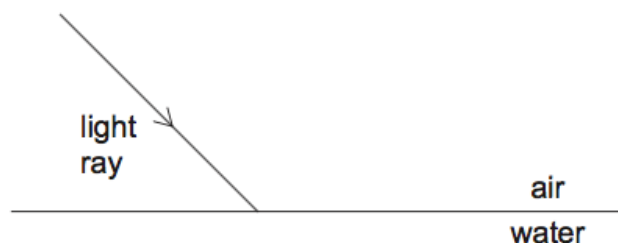


Markscheme

B

84. A light ray passes from air to water as shown.

[1 mark]



What are the change in the wavelength of the light wave and the change in the angle the ray makes with the normal to the surface?

	Wavelength	Angle with normal
A.	increases	increases
B.	increases	decreases
C.	decreases	increases
D.	decreases	decreases

Markscheme

D